

PREETHAM BHAVIRISETTY

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building scalable, production-grade backend systems using Python, Node.js, and TypeScript. Specialized in designing RESTful APIs, developing LLM-powered applications with RAG pipelines, vector search, and AI orchestration frameworks. Hands-on expertise with AWS managed services (ECS, Lambda, SageMaker, Aurora, OpenSearch), CI/CD automation, Docker/Kubernetes containerization, and infrastructure as code. Proven ability to deliver reliable, fault-tolerant distributed systems in fast-paced Agile environments with a strong focus on performance, observability, and maintainability.

PROFESSIONAL EXPERIENCE

Software Engineer – Backend & AI | Aetna (CVS Health)

Sep 2025 – Present

- Architected and scaled YAVA, a conversational AI platform built on Node.js and TypeScript, handling complex healthcare workflows with low-latency, highly available backend services deployed on AWS ECS (Elastic Container Service) for container orchestration and auto-scaling.
- Integrated IBM Watson Assistant for NLU-driven intent recognition and X Orchestrator for multi-agent coordination, enabling deterministic workflow routing, graceful fallbacks, and safe agent handoffs across conversational flows.
- Built an LLM control layer with rule-based policy enforcement and contextual memory enrichment using AWS DynamoDB as the session store, reducing hallucinations and ensuring regulatory compliance for PHI-sensitive healthcare interactions.
- Leveraged AWS Lambda for event-driven processing of asynchronous AI tasks (e.g., summarization, post-call analysis), decoupling heavy workloads from the real-time request path and improving overall system responsiveness.
- Implemented structured observability using AWS CloudWatch Logs, correlation IDs, distributed tracing via OpenTelemetry, and custom metrics dashboards - cutting mean time to detect (MTTD) incidents by ~40%.
- Designed multi-channel backend APIs for voice-to-chat/SMS switching by integrating external communication platforms SFMC, ensuring high delivery success rates and end-to-end message traceability.
- Owned CI/CD pipelines with Jenkins and Octopus Deploy, combined with environment-specific configuration management via Azure App Config, enabling zero-downtime deployments across dev, staging, and production environments.

Software Development Engineer | Cloud5.ai

Dec 2024 – Aug 2025

- Developed scalable backend microservices in Node.js (TypeScript) and Python (FastAPI), exposing robust REST APIs that handled 10,000+ daily transactions with 99.9% uptime, deployed on AWS ECS with Aurora PostgreSQL as the primary relational store.
- Designed and implemented RAG pipelines using Python, LangChain, and AWS OpenSearch (vector search), enabling semantic search across large document corpora and significantly reducing manual lookup effort for operations teams.
- Built LLM-powered microservices for prompt orchestration, context assembly, and structured response validation using OpenAI and Anthropic model APIs, increasing automation coverage and reducing manual intervention in user workflows by ~35%.
- Optimized API performance through Redis caching (AWS ElastiCache), efficient PostgreSQL query tuning with `pg_explain`, and Celery-based async task queues - reducing P99 latency by ~30% during peak traffic.
- Integrated scikit-learn and AWS SageMaker ML models into backend services for real-time scoring and anomaly detection on financial and operational data, enabling faster data-driven decision-making.
- Built data pipelines using AWS S3 and Lambda triggers to ingest, transform, and store structured/unstructured data into MongoDB Atlas and PostgreSQL, supporting both operational and analytics workloads.
- Automated CI/CD workflows using GitHub Actions with Docker builds, ECR image pushes, and ECS deployments - cutting manual deployment effort by 50% and ensuring consistent, repeatable releases.

Software Developer | CRED, India

Jan 2020 – Jul 2022

- Developed internal automation microservices using Java (Spring Boot) and Python (FastAPI) to process high-volume credit approval requests, reducing processing time by ~40%.
- Integrated Redis Streams and Apache Kafka for real-time task queuing and event-driven data ingestion, improving system throughput and decoupling service dependencies.
- Embedded Python-based ML models into Spring Boot services for automated approval scoring, enabling real-time credit decisioning and cutting manual review workload significantly.
- Configured end-to-end CI/CD pipelines with Jenkins and Docker for both Java and Python services, improving deployment reliability and simplifying rollback processes during incidents.
- Implemented API-level logging and monitoring using Prometheus metrics and the ELK Stack (Elasticsearch, Logstash, Kibana), enabling proactive alerting and real-time issue detection.
- Collaborated with cross-functional teams (QA, frontend, business stakeholders) in Agile sprints to deliver tested backend features from requirements to production release.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, Bash

Frameworks & Libraries: FastAPI, Node.js, Spring Boot, React.js, Tailwind CSS, LangChain, scikit-learn

AI / ML: LLM APIs (OpenAI, Anthropic), RAG Pipelines, Vector Search, AWS SageMaker, IBM Watson Assistant, Prompt Engineering

Databases: PostgreSQL, MongoDB, MySQL, Cassandra, Redis, DynamoDB, OpenSearch (vector store)

AWS Services: ECS, Lambda, S3, SageMaker, Aurora, ElastiCache, OpenSearch, Fargate, CloudWatch, ECR

DevOps & Cloud: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Azure App Config, Octopus Deploy

API & Integration: REST APIs, OpenAPI/Swagger, OAuth2, JWT, Apache Kafka, Redis Streams

Observability: Prometheus, Grafana, ELK Stack, AWS CloudWatch, OpenTelemetry

Methodologies: Agile (Scrum), TDD, Microservices, Domain-Driven Design, Clean Architecture

PROJECTS

NoteLite — Cloud-Native Note-Taking Platform notelite.org

Built a cloud-native note-taking app with a Node.js/TypeScript REST API backend, React.js + Tailwind CSS frontend, and PostgreSQL for persistent storage - containerized with Docker and deployed on AWS. Implemented token-based authentication (JWT/OAuth2) with user-level data isolation, and optimized read/write performance for high-frequency note operations.

AI-Powered Finance Tracker [Live Demo](#)

Developed a full-stack finance tracking platform with Python/Node.js backend APIs, React.js + Tailwind CSS frontend, and MongoDB for flexible multi-profile financial data storage on AWS. Integrated LLM-based summarization (OpenAI API) to auto-generate natural-language financial insights - monthly spend summaries, category trend analysis, and anomaly flagging - reducing manual analysis time for users.

EDUCATION

Master of Science in Computer Science — Binghamton University, New York

Bachelor of Science in Information Technology — VR Siddhartha Engineering College, India

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate | [credly.com/badges](https://www.credly.com/badges)
- AWS Certified AI Practitioner | [credly.com/badges](https://www.credly.com/badges)